

PYTHON PRACTICAL 8

ROLL NO : S054	NAME : KOMAL YADAV
CLASS : SYIT	SUBMISSION : 11/08/26

Q. 1. Write algorithm for the following:

a. Write a program to accept and pass radius to a function that returns area and circumference (using tuple).

CODE :

```
import math
#Function to calculate area and circumference
def circle_details(r):
    area = math.pi * r * r
    circum = 2 * math.pi * r
    return (area, circum)    #Returning as a tuple

#Accept radius from the user
radius = float(input("Enter radius: "))

#Call Function
result = circle_details(radius)

#Display results
print("Area: ", result[0])
print("Circumference: ", result[1])
print("KOMAL S049")
```

OUTPUT :

```
===== RESTART: C:/Users/info/O
Enter radius: 3
Area:  28.274333882308138
Circumference:  18.84955592153876
KOMAL S049
```

b. Write a program to perform basic file operations on text files and binary files.

CODE :

1)

```
#Write to a text file
with open("example.txt", "w") as f:
    #The with statement ensures the file will automatically close after write/read
    f.write("Hello Students!\n Welcome to Python File Handling.")

# Read from text file
with open("example.txt", "r") as f:
    content = f.read()
    print("Text File contents:\n", content)
print("KOMAL S049")
```

2)

```
#Write binary data(bytes) to a File
with open("b_sample.bin", "wb") as bf:
    data = b"This is binary data"
    bf.write(data)

# Read binary data from the file
with open("b_sample.bin", "rb") as bf:
    b_content = bf.read()
    print("Binary File contents:\n", b_content)
print("KOMAL S049")
```

OUTPUT :

1)

```
-----
Text File contents:
Hello Students!
Welcome to Python File Handling.
KOMAL S049
|
```

2)

```
-----
Binary File contents:
b'This is binary data'
KOMAL S049
|
```

c. Write a Python program to read last n lines of a file.

CODE :

```
def read_last_n_lines(filename, n):
    with open(filename, "r") as f:
        lines = f.readlines() # Reads all lines from the file into a list called
                               # lines. So - lines = ["Line 1\n", "Line 2\n", "Line 3\n"]
        return lines[-n:]    # Get last n lines. Negative indexing (-n:)
                               # is used to slice the list from the end.
                               # Example: ["Line 2\n", "Line 3\n"]

# File name and n value
filename = "example.txt"
n = int(input("Enter number of last lines to read: "))

# Read and print lines
# Executes the function and stores the result in last_lines
last_lines = read_last_n_lines(filename, n)

print("\nLast", n, "Lines:")
for line in last_lines:
    print(line.strip()) # Removes extra spaces and newline characters (\n)
                       # Prints each line cleanly without blank lines.

print("KOMAL S049")
```

OUTPUT :

```
===== RESIAKI: C:/Users/info/OneDri'
Enter number of last lines to read: 2

Last 2 Lines:
Hello Students!
Welcome to Python File Handling.
KOMAL S049
```